

SHARED ATTRIBUTE SUPERVISION FOR RESPIRATORY SOUND LEARNING ACROSS NATIVE TASKS

ABSTRACT

Respiratory-sound datasets describe related events with different annotation units and label granularity. We study how sharing their supervision affects native-task performance and fine-attribute discrimination. A shared BEATs encoder learns abnormality, crackle, and wheeze from ICBHI and SPRSound, with readouts for their four-class and binary tasks. Under ICBHI-test-based checkpoint selection, the joint model obtains an ICBHI Score of $61.17 \pm 0.31\%$ and a SPRSound official Score of $90.70 \pm 0.34\%$; the published ICBHI-only PAFA reference is $64.84 \pm 0.60\%$. HF supports an additional-supervision study and attribute evaluation; KAUH tests external patient-level transfer. Separate validation-selected controls distinguish source sharing and annotation information from classification design, measuring native-task retention and the effect of the final readout.

Index Terms— respiratory sounds, shared supervision, heterogeneous annotations, acoustic attributes, hierarchical decoding

1. INTRODUCTION

Automated respiratory-sound analysis aims to recognize adventitious events reliably and support quantitative auscultation [1, 2]. Scarce annotations and variation across patients and recording devices complicate this task. Learning from related datasets offers additional supervision, provided that differences in their annotations and task definitions are handled explicitly.

Existing methods improve data efficiency through augmentation and contrastive learning, as illustrated by RespireNet and Patch-Mix [3, 4]. SG-SCL and PAFA address variation associated with devices and patients [5, 6]. BEATs, OPERA, and respiratory audio pretraining provide reusable representations, whose downstream utility also depends on the adaptation setting [7–10].

A central difficulty is that related acoustic events are annotated for different objectives. ICBHI labels respiratory cycles by crackle and wheeze occurrence, whereas SPRSound supplies fine event annotations alongside a binary evaluation endpoint [1, 11]. HF Lung provides timed phase and event labels, and KAUH uses recording-level sound categories [2, 12]. A model trained only for a coarse endpoint may leave avail-

able fine annotations unused. Conversely, treating an unsupported attribute as absent can introduce unjustified negative supervision. Thus, sharing data requires separating the information available for training from the labels required at evaluation.

Related studies address source reuse and label structure. SPRSound includes data-fusion experiments, and mixed-set training combines lung and tracheal recordings [11, 13]. LungMix aligns labels for cross-dataset generalization, while BTS-CARD addresses metadata-related bias [14, 15]. PC-MCL studies multi-label composition for concatenated cycles, and OCAD uses structured attribute-to-class decoding [16, 17]. These settings emphasize fusion, source generalization, or label composition; they do not directly isolate supervision and readout effects when cycle and event datasets retain different native endpoints.

More generally, universal taxonomies and availability-aware losses support learning with overlapping or heterogeneous labels [18, 19]. DCASE combines heterogeneous audio annotations through class mapping, masking, and dataset-specific evaluation [20]. These mechanisms provide a basis for sharing supervision. For respiratory cycle and event tasks, the specific question is which fine acoustic attributes remain useful across annotation conditions, and how their contribution to native-task performance depends on the classification interface. Comparing a model using fine attribute labels with a native binary classifier using only coarse labels changes both factors at once.

We formulate this problem as learning shared respiratory attributes under explicit annotation availability while retaining native task definitions. A shared BEATs encoder learns abnormality, crackle, and wheeze from ICBHI and SPRSound, with two-level readouts for their four-class and binary tasks. HF supports a separate auxiliary-supervision study and attribute evaluation; KAUH provides external patient-level assessment. Controlled comparisons vary the training sources, available attribute labels, and classification or readout design to examine native-task retention, attribute reuse, and the respective roles of supervision information and task decisions.

Figure 1 placeholder

Sample-level acoustic characteristics across
ICBHI, SPRSound, HF Lung, and KAUH

Fig. 1. Acoustic cues and annotation granularity across datasets. Figure and analysis pending.

2. DATASETS AND ACOUSTIC CHARACTERISTICS

2.1. Datasets and Study Roles

Core training uses ICBHI respiratory cycles and SPRSound events. ICBHI labels cycles as Normal, Crackle, Wheeze, or Both and provides 4,142/2,756 training/test cycles in its official recording split [1]; this split is not strictly patient-held-out. Both denotes C/W co-presence within a cycle. SPRSound provides seven event categories, while its native BioCAS2022 Task1-1 evaluates Normal/Adventitious classification on 1,429 official inter-subject test events [11].

HF_Lung_V1 contains 9,765 15-s recordings, split into 7,809 source-training and 1,956 source-test recordings, with timed phase and adventitious-sound annotations [2]. These distinguish inhalation/exhalation (I/E), discontinuous adventitious sounds (DAS, source label D), and continuous adventitious sounds (CAS: Wheeze, Rhonchi, and Stridor). Source-training data support a separate auxiliary-supervision study; source-test data evaluate the shared attributes. KAUH supplies external evaluation: 336 files from 112 subjects, with Bell, Diaphragm, and Extended (B/D/E) versions of each recording [12]. These versions are aggregated within patient.

2.2. Acoustic Cues and Annotation Granularity

[Class-matched acoustic cues and annotation-granularity comparison to accompany Figure 1.]

2.3. Annotation Alignment and Input Preparation

Shared core targets (a, c, w) encode abnormality, Crackle, and Wheeze. ICBHI labels specify all three; compatible SPRSound labels use the same decomposition, merging Coarse and Fine Crackle. Rhonchi/Stridor provide only abnormality supervision, and unsupported core targets are masked. HF supplies separate C/W auxiliary supervision, with D mapped to Crackle and no abnormality target.

For KAUH, N, C/I C, E W/I E W, and I C E W map to Normal, Crackle, Wheeze, and Both, respectively. Crep,

Bronchial, and I C B are excluded, leaving 86 compatible patients. This evaluation uses sound labels rather than disease diagnoses or the complete nine-class taxonomy.

Inputs are mono at 16 kHz. ICBHI cycles, SPRSound events, and KAUH files use 5 s through front truncation or repetition with boundary fading. HF uses three continuous windows: [0,5], [5,10], and [10,15] s. Native sample identities are retained. ICBHI, SPRSound, and KAUH retain patient identifiers; HF uses recording-date proxies, not patient IDs. Split and selection details are given in Section 4.

3. METHOD

Our method connects heterogeneous annotations through shared acoustic attributes, then recovers native task labels from their predictions (Figure 2). This separates the information used to learn the representation from the rules used to make task decisions.

3.1. Shared Acoustic Attribute Predictors

For waveform x_i , a BEATs iter3+ AS2M encoder [7] pre-trained on AudioSet [21] produces 768-dimensional token features. Mean pooling followed by a linear projection gives $h_i \in \mathbb{R}^{256}$. Three parallel linear heads predict abnormality (A), Crackle (C), and Wheeze (W):

$$\begin{aligned} p_{ia} &= \text{softmax}(W_a h_i + b_a)_{\text{Abnormal}}, \\ p_{ic} &= \sigma(W_c h_i + b_c), \quad p_{iw} = \sigma(W_w h_i + b_w). \end{aligned} \quad (1)$$

A uses two logits, whereas C/W each use one. All heads jointly fine-tune the encoder on ICBHI and SPRSound. C/W are marginal attribute probabilities; the two-level hierarchy is applied in the decision readout.

3.2. Learning from Available Annotations

Partial-label learning distinguishes unobserved targets from observed negatives [22,23]. The alignment in Section 2.3 supplies targets y_{ik} and availability masks m_{ik} for $k \in \{a, c, w\}$. A mask is one only when the source annotation supports that

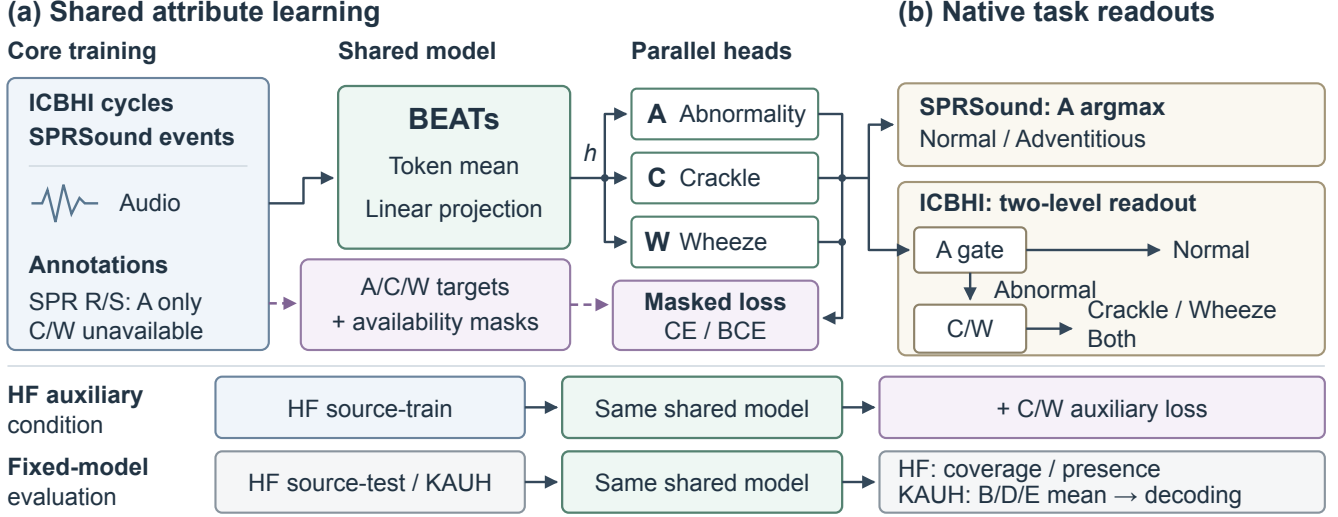


Fig. 2. Shared attribute learning and native task readouts. (a) ICBHI and SPRSound annotations supervise available A/C/W targets, while their audio trains a common encoder and three parallel heads. (b) The same scores support binary and two-level four-class decisions. HF source-train supplies a separate C/W auxiliary-supervision condition; HF source-test and KAUH evaluate fixed models.

target. For batch B , let $B_k = \{i \in B : m_{ik} = 1\}$ and $K_B = \{k : |B_k| > 0\}$. The classification objective averages available supervision within each node:

$$\mathcal{L}_{\text{cls}}(B) = \frac{1}{|K_B|} \sum_{k \in K_B} \frac{1}{|B_k|} \sum_{i \in B_k} \ell_k(p_{ik}, y_{ik}), \quad (2)$$

where ℓ_a is two-class cross-entropy and ℓ_c, ℓ_w are binary cross-entropy. This lets compatible labels share a predictor while masking unsupported targets. Dataset-homogeneous batches contribute equal numbers of updates per epoch.

To regularize patient-specific variation, we retain PAFA’s patient cohesion-separation and global patient alignment objectives [6]:

$$\mathcal{L}_{\text{core}} = \mathcal{L}_{\text{cls}} + 50\mathcal{L}_{\text{PCSL}} + 0.0005\mathcal{L}_{\text{GPAL}}. \quad (3)$$

A separate attention-based projection of encoder tokens supplies their features during training; patient identifiers are qualified by dataset.

3.3. Two-Level Native Task Readouts

The Normal/Abnormal argmax defines $\hat{a}$ (ties select Normal), and $\hat{c} = \mathbb{I}[p_c \geq \tau_c]$, $\hat{w} = \mathbb{I}[p_w \geq \tau_w]$ define attribute decisions. SPRSound uses $\hat{a}$ for Normal/Adventitious classification. ICBHI first assigns Normal when $\hat{a} = 0$; otherwise, two positive attributes yield Both and a single positive attribute yields its class. If neither is positive, the larger $p_k - \tau_k$ margin selects Crackle or Wheeze, with ties assigned to Crackle.

Shared thresholds independently maximize the equal-source mean of available-target binary F1 on core validation

data. Candidates are the observed probabilities plus 0 and 1; ties select the larger threshold. Changing the readout while fixing the encoder and attribute scores isolates decision-rule effects from changes in the learned representation.

3.4. HF Auxiliary-Supervision Condition

To examine additional attribute supervision, we retain the architecture and add HF source-training windows:

$$\mathcal{L}_{\text{HF-on}} = \mathcal{L}_{\text{core}} + 0.25\mathcal{L}_{\text{HF}}, \quad (4)$$

using one batch of 32 HF windows per core update. A window overlapping D, Wheeze, Rhonchi, or Stridor supplies both C/W targets: each is one when the corresponding D or Wheeze annotation overlaps, and zero otherwise. $\mathcal{L}_{\text{HF}}$ averages their node-wise BCE losses. Gap, empty, and phase-only windows are masked; HF supplies no A target. The derived negatives assume that an absent attribute annotation denotes absence within these eligible windows, although uncertain events may be omitted in the source annotations [2]. HF is excluded from threshold fitting, checkpoint selection, and early stopping; its source-test partition is reserved for evaluation.

4. EVALUATION

We evaluate whether shared attribute supervision supports both native classification tasks, how additional HF supervision changes performance, and which effects arise from supervision information or classification design. Tables 1–2 address these questions while retaining the four dataset roles in Section 2.

Table 1. Native-task benchmark results and frozen-encoder context (%). Published ICBHI-only rows retain their reported mean $\pm$ SD; our joint model uses three runs with ICBHI-test-based checkpoint selection. The lower block contains reported single-seed, single-dataset frozen references. Dashes denote unreported endpoints.

Model	ICBHI four-class [1]			SPRSound inter binary [11]		
	Sp	Se	Score	AS	HS	Official Score
<i>Published ICBHI-only benchmarks</i>						
Patch-Mix CL [4]	81.66 ± 3.83	43.07 ± 2.80	62.37 ± 0.61	—	—	—
SG-SCL [5]	79.87 ± 8.89	43.55 ± 5.93	61.71 ± 1.61	—	—	—
PAFA [6]	82.05 ± 1.95	47.63 ± 2.23	64.84 ± 0.60	—	—	—
PC-MCL [16]	79.04 ± 1.90	51.71 ± 2.98	65.37 ± 0.73	—	—	—
Ours	74.98 ± 5.69	47.35 ± 5.35	61.17 ± 0.31	90.95 ± 0.31	90.45 ± 0.39	90.70 ± 0.34
<i>Frozen single-dataset context: seed 42</i>						
AST [24]	55.79	46.64	51.22	90.14	89.83	89.99
BEATs [7]	56.17	46.56	51.37	92.08	91.92	92.00
PANNs [25]	56.62	47.32	51.97	91.37	91.24	91.31
OPERA-CT [8]	39.58	37.64	38.61	79.48	79.42	79.45
HeAR [26]	66.69	31.27	48.98	88.16	88.09	88.12

4.1. Experimental Protocol and Metrics

The shared model uses the 5-s inputs and losses in Section 3, batch size 32, Adam with learning rate 5×10^{-5} and weight decay 10^{-6} , a 50-epoch cosine schedule, and parameter EMA 0.5. Each joint training epoch contains 326 updates, equally divided between ICBHI and SPRSound.

For our benchmark in Table 1, checkpoints and patience-10 early stopping use ICBHI official-test Score. This follows the test-based selection approach in the released PAFA, SG-SCL, and Patch-Mix implementations [4–6]. C/W thresholds use internal validation data; SPRSound inter is evaluated once per selected checkpoint. We report three completed runs with seeds 0, 1, and 42, using their mean and sample standard deviation.

The controlled comparisons in Table 2 instead use patient-grouped validation subsets of official training data for checkpoint selection, early stopping, and thresholds. Test sets are accessed after selection. These comparisons use seeds 0, 1, and 42; their joint reference is trained under this validation protocol and is distinct from the benchmark model in Table 1.

ICBHI Score is $(Sp + Se)/2$, where Sp is Normal recall and Se counts abnormal cycles assigned to the correct abnormal subclass. SPRSound Task1-1 uses the official inter-subject test and $Score = (AS + HS)/2$, with $AS = (Se + Sp)/2$ and $HS = 2SeSp/(Se + Sp)$; here Se and Sp are Adventitious and Normal recall. Macro-F1 averages the class-wise F1 values. On the HF source-test partition, interval recall counts a labeled D/W interval when any overlapping 5-s window reaches the fixed attribute threshold. Recording AUROC uses maximum scores across the three windows on recordings with at least one D/W/R/S annotation; absent target annotations supply derived negatives. KAUH averages B/D/E probabilities within patient before reporting binary balanced

accuracy (BA), four-class Macro-F1, and unweighted average recall (UAR) on the 86 compatible patients.

4.2. Benchmark Results and Comparison

Table 1 compares our joint model with published ICBHI-only methods. Our model achieves ICBHI Score $61.17 \pm 0.31\%$ and SPRSound official Score $90.70 \pm 0.34\%$. Relative to PAFA, our mean Se differs by only 0.28 percentage points (pp) (47.35% versus 47.63%), whereas Sp is 7.07 pp lower (74.98% versus 82.05%). The ICBHI Score difference is therefore concentrated in specificity. Our ICBHI Score is 3.67 pp below PAFA and 4.20 pp below PC-MCL. Table 2 quantifies native-task retention using matched single-source and joint models.

Niizumi et al. [10] evaluated frozen audio encoders with trainable task networks, reporting performance differences across tasks and head architectures. The lower block of Table 1 provides a separate set of single-run references using 2-s windows. In this setting, frozen BEATs reaches 51.37% ICBHI Score and 92.00% SPRSound Score. These references train separate native heads and provide context for the full-fine-tuning controls.

4.3. Ablations of Supervision and Classification

Table 2 compares source sharing, annotation information, classification design, and HF auxiliary supervision. The full joint model without HF is the common reference. Retrained variants retain the BEATs initialization, input processing, PAFA regularization, optimizer, maximum update budget, and validation cadence. Single-source variants use only their own source for fitting, checkpoint selection, and calibration. C/W attribute AUROC is the mean of the two binary AU-

ROCs on SPRSound inter events whose original annotations support the corresponding targets.

Table 2. Validation-selected ablations (% , mean $\pm$ SD). HF-off is the full-model reference; only Full model + HF adds HF supervision. Direct C/W thresholding changes only the ICBHI conversion rule at fixed scores and thresholds; dashes omit unchanged outputs. SPR Score denotes official Score.

Variant	ICBHI Score	SPR Score	SPR C/W AUROC
Full model (HF-off)	xxx $\pm$ xxx	xxx $\pm$ xxx	xxx $\pm$ xxx
Full model + HF	xxx $\pm$ xxx	xxx $\pm$ xxx	xxx $\pm$ xxx
ICBHI only	xxx $\pm$ xxx	xxx $\pm$ xxx	xxx $\pm$ xxx
SPRSound only	xxx $\pm$ xxx	xxx $\pm$ xxx	xxx $\pm$ xxx
Coarse SPR labels	xxx $\pm$ xxx	xxx $\pm$ xxx	xxx $\pm$ xxx
Native + attributes	xxx $\pm$ xxx	xxx $\pm$ xxx	xxx $\pm$ xxx
Direct C/W thresholding [16]	xxx $\pm$ xxx	—	—

Source sharing and annotation information. Joint versus single-source training changes ICBHI and SPRSound native Scores by xxx and xxx pp. Withholding SPR C/W targets while retaining its audio and abnormality labels changes attribute AUROC by xxx pp. The retained loss weights remain fixed, and withheld fine labels are used only for evaluation. This comparison tests fine-attribute reuse when the target source provides coarse supervision.

Classification design. Native heads with matched attribute supervision change ICBHI/SPRSound Scores by xxx/xxx pp relative to shared-attribute readouts. Direct C/W thresholding fixes our model’s scores and thresholds and replaces only the ICBHI two-level rule with C/W-only decoding: neither attribute gives Normal, one gives its class, and two give Both. PC-MCL adds an explicit Normal target during multi-cycle training, while its inference also uses this discrete C/W-to-class conversion [16]. Our comparison keeps the trained representation and attribute supervision fixed. ICBHI Score and Both recall change by xxx/xxx pp, while attribute AUROCs remain unchanged, isolating the effect of the final conversion rule.

Additional HF supervision and external evaluation. Full model + HF adds the objective in Section 3.4 to the HF-off reference, keeping core data, seeds, optimizer, and core-update budget fixed. ICBHI/SPRSound Scores change by xxx/xxx pp, while ICBHI abnormal-subclass recall changes from xxx% to xxx%. On HF source-test, D/W recording AUROCs change from xxx/xxx% to xxx/xxx%, and interval recalls from xxx/xxx% to xxx/xxx%. KAUH binary BA and four-class Macro-F1 change from xxx/xxx% to xxx/xxx%, with UAR changing from xxx% to xxx%. These endpoints assess attribute-specific benefits and costs across native and external tasks.

5. CONCLUSION

We presented annotation-aware shared-attribute learning for respiratory-sound datasets with heterogeneous labels. A shared encoder learns abnormality, crackle, and wheeze from ICBHI cycles and SPRSound events, while two-level readouts preserve their native four-class and binary tasks. HF provides an auxiliary-supervision study and attribute evaluation; KAUH assesses external transfer at the patient level. Matched comparisons separate source sharing and annotation granularity from classification design, connecting fine-attribute discrimination to native-task performance. This separation allows shared supervision to be assessed through both the acoustic information learned and the benefits and costs of converting that information into each dataset’s task decisions.

6. REFERENCES

- [1] Bruno M. Rocha, Dimitris Filos, Luís Mendes, Gorkem Serbes, Sezer Ulukaya, Yasemin P. Kahya, Nikša Jakovljevic, Tatjana L. Turukalo, Ioannis M. Vogiatzis, Eleni Perantoni, Evangelos Kaimakamis, Pantelis Natsiavas, Ana Oliveira, Cristina Jácome, Alda Marques, Nicos Maglaveras, Rui Pedro Paiva, Ioanna Chouvarda, and Paulo de Carvalho, “An open access database for the evaluation of respiratory sound classification algorithms,” *Physiological Measurement*, vol. 40, no. 3, pp. 035001, 2019.
- [2] Fu-Shun Hsu, Shang-Ran Huang, Chien-Wen Huang, Chao-Jung Huang, Yuan-Ren Cheng, Chun-Chieh Chen, Jack Hsiao, Chung-Wei Chen, Li-Chin Chen, Yen-Chun Lai, Bi-Fang Hsu, Nian-Jhen Lin, Wan-Ling Tsai, Yi-Lin Wu, Tzu-Ling Tseng, Ching-Ting Tseng, Yi-Tsun Chen, and Feipei Lai, “Benchmarking of eight recurrent neural network variants for breath phase and adventitious sound detection on a self-developed open-access lung sound database—HF_Lung_V1,” *PLOS ONE*, vol. 16, no. 7, pp. e0254134, 2021.
- [3] Siddhartha Gairola, Francis Tom, Nipun Kwatra, and Mohit Jain, “RespireNet: A deep neural network for accurately detecting abnormal lung sounds in limited data setting,” in *2021 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2021, pp. 527–530.
- [4] Sangmin Bae, June-Woo Kim, Won-Yang Cho, Hyerim Baek, Soyoun Son, Byungjo Lee, Changwan Ha, Kyongpil Tae, Sungnyun Kim, and Se-Young Yun, “Patchmix contrastive learning with audio spectrogram transformer on respiratory sound classification,” in *Inter-speech 2023*, 2023, pp. 5436–5440.

- [5] June-Woo Kim, Sangmin Bae, Won-Yang Cho, Byungjo Lee, and Ho-Young Jung, "Stethoscope-guided supervised contrastive learning for cross-domain adaptation on respiratory sound classification," in *ICASSP 2024—2024 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2024, pp. 1431–1435.
- [6] Seung Gyu Jeong and Seong Eun Kim, "Patient-aware feature alignment for robust lung sound classification: Cohesion-separation and global alignment losses," in *Interspeech 2025*, 2025, pp. 1018–1022.
- [7] Sanyuan Chen, Yu Wu, Chengyi Wang, Shujie Liu, Daniel Tompkins, Zhuo Chen, Wanxiang Che, Xiangzhan Yu, and Furu Wei, "BEATs: Audio pre-training with acoustic tokenizers," in *Proceedings of the 40th International Conference on Machine Learning*. 2023, vol. 202 of *Proceedings of Machine Learning Research*, pp. 5178–5193, PMLR.
- [8] Yuwei Zhang, Tong Xia, Jing Han, Yu Yvonne Wu, Georgios Rizos, Yang Liu, Mohammed Mosuily, Jagmohan Chauhan, and Cecilia Mascolo, "Towards open respiratory acoustic foundation models: Pretraining and benchmarking," in *Advances in Neural Information Processing Systems*, 2024, vol. 37, pp. 27024–27055.
- [9] Daisuke Niizumi, Daiki Takeuchi, Masahiro Yasuda, Binh Thien Nguyen, Yasunori Ohishi, and Noboru Harada, "Towards pre-training an effective respiratory audio foundation model," in *Interspeech 2025*, 2025, pp. 998–1002.
- [10] Daisuke Niizumi, Daiki Takeuchi, Masahiro Yasuda, Binh Thien Nguyen, Yasunori Ohishi, and Noboru Harada, "Assessing the utility of audio foundation models for heart and respiratory sound analysis," in *2025 47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2025, pp. 1–4.
- [11] Qing Zhang, Jing Zhang, Jiajun Yuan, Huajie Huang, Yuhang Zhang, Baoqin Zhang, Gaomei Lv, Shuzhu Lin, Na Wang, Xin Liu, Mingyu Tang, Yahua Wang, Hui Ma, Lu Liu, Shuhua Yuan, Hongyuan Zhou, Jian Zhao, Yongfu Li, Yong Yin, Liebin Zhao, Guoxing Wang, and Yong Lian, "SPRSound: Open-source SJTU paediatric respiratory sound database," *IEEE Transactions on Biomedical Circuits and Systems*, vol. 16, no. 5, pp. 867–881, 2022.
- [12] Mohammad Fraiwan, Luay Fraiwan, Basheer Khasawneh, and Ali Ibnian, "A dataset of lung sounds recorded from the chest wall using an electronic stethoscope," *Data in Brief*, vol. 35, pp. 106913, 2021.
- [13] Fu-Shun Hsu, Shang-Ran Huang, Chang-Fu Su, Chien-Wen Huang, Yuan-Ren Cheng, Chun-Chieh Chen, Chun-Yu Wu, Chung-Wei Chen, Yen-Chun Lai, Tang-Wei Cheng, Nian-Jhen Lin, Wan-Ling Tsai, Ching-Shiang Lu, Chuan Chen, and Feipei Lai, "A dual-purpose deep learning model for auscultated lung and tracheal sound analysis based on mixed set training," *Biomedical Signal Processing and Control*, vol. 86, pp. 105222, 2023.
- [14] Shijia Ge, Weixiang Zhang, Shuzhao Xie, Baixu Yan, and Zhi Wang, "LungMix: A mixup-based strategy for generalization in respiratory sound classification," in *ICASSP 2025—2025 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2025, pp. 1–5.
- [15] Heejoon Koo, Miika Toikkanen, Yoon Tae Kim, Soo Yong Kim, and June-Woo Kim, "Empowering multimodal respiratory sound classification with counterfactual adversarial debiasing for out-of-distribution robustness," in *ICASSP 2026—2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026.
- [16] Seung Gyu Jeong and Seong-Eun Kim, "PC-MCL: Patient-consistent multi-cycle learning with multi-label bias correction for respiratory sound classification," in *ICASSP 2026—2026 IEEE International Conference on Acoustics, Speech and Signal Processing*, 2026, pp. 2665–2669.
- [17] Xinyu Zhang, Wei Zhao, and Haicheng Liang, "OCAD: Overlap composition and class-atom decoding for respiratory-sound classification," *Applied Sciences*, vol. 16, no. 13, pp. 6832, 2026.
- [18] Petra Bevandić, Marin Oršić, Ivan Grubišić, Josip Šarić, and Siniša Šegvić, "Multi-domain semantic segmentation with overlapping labels," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022, pp. 2615–2624.
- [19] Mark Schutera, Luca Rettenberger, Christian Pylatiuk, and Markus Reischl, "Methods for the frugal labeler: Multi-class semantic segmentation on heterogeneous labels," *PLOS ONE*, vol. 17, no. 2, pp. e0263656, 2022.
- [20] Samuele Cornell, Janek Ebberts, Constance Douwes, Irene Martín-Morató, Manu Harju, Annamaria Mesaros, and Romain Serizel, "DCASE 2024 task 4: Sound event detection with heterogeneous data and missing labels," arXiv:2406.08056, 2024.
- [21] Jort F. Gemmeke, Daniel P. W. Ellis, Dylan Freedman, Aren Jansen, Wade Lawrence, R. Channing Moore, Manoj Plakal, and Marvin Ritter, "Audio Set: An ontology and human-labeled dataset for audio events,"

in *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 776–780.

- [22] Gonglei Shi, Li Xiao, Yang Chen, and S. Kevin Zhou, “Marginal loss and exclusion loss for partially supervised multi-organ segmentation,” *Medical Image Analysis*, vol. 70, pp. 101979, 2021.
- [23] Donghao Zhou, Pengfei Chen, Qiong Wang, Guangyong Chen, and Pheng-Ann Heng, “Acknowledging the unknown for multi-label learning with single positive labels,” in *Computer Vision—ECCV 2022*. 2022, vol. 13684 of *Lecture Notes in Computer Science*, pp. 423–440, Springer.
- [24] Yuan Gong, Yu-An Chung, and James Glass, “AST: Audio spectrogram transformer,” in *Interspeech 2021*, 2021, pp. 571–575.
- [25] Qiuqiang Kong, Yin Cao, Turab Iqbal, Yuxuan Wang, Wenwu Wang, and Mark D. Plumbley, “PANNs: Large-scale pretrained audio neural networks for audio pattern recognition,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 28, pp. 2880–2894, 2020.
- [26] Sebastien Baur, Zaid Nabulsi, Wei-Hung Weng, Jake Garrison, Louis Blankemeier, Sam Fishman, Christina Chen, Sujay Kakarmath, Minyoi Maimbolwa, Nsala Sanjase, Brian Shuma, Yossi Matias, Greg S. Corrado, Shwetak Patel, Shravya Shetty, Shruthi Prabhakara, Monde Muyoyeta, and Diego Ardila, “HeAR—health acoustic representations,” arXiv:2403.02522, 2024.